

Siddharth Arora

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SUMMARY

Senior ML/AI Engineer with experience owning production machine learning systems across forecasting, semantic search, NLP, and applied AI. Strong track record of turning ambiguous business problems into scalable ML solutions through experimentation, system design, evaluation, and cross-functional execution.

TECHNICAL SKILLS

Languages: Python (pandas, numpy, polars, pytorch), R, SQL

ML & AI: Hugging Face, Pydantic, LangGraph, LangChain, LangFuse, OpenAI APIs, Google Vertex AI

Cloud & Infrastructure: AWS (EC2, ECR, S3, Batch, SNS, Lambda), Docker, FastAPI, Metaflow, OuterBounds, CI/CD (CircleCI)

Data & Platforms: Snowflake, Git

Domains: Time Series Forecasting, Semantic Search, Image Tagging, Classification, Regression, NLP, AI Evaluation

EXPERIENCE

Senior Machine Learning Engineer • [Realtor.com](#) • Remote (Toronto, Canada) *Jun. 2021 – Present*
Consumer real estate platform powering discovery, search, and lead generation.

- Owned the design and productionization of a large-scale time-series forecasting system for monthly leads across US zip codes, improving accuracy by **30%** while reducing runtime and system complexity; details at techblog.realtor.com.
- Built an **AI/NLP ensemble** powering Realtor.com's home search bar, replacing regex-based query parsing with learned intent extraction to convert free-text queries into structured filters, improving parsing **F1 by 36%** and reducing **p95 latency by 50%**, while contributing to downstream gains in user engagement and search quality.
- Optimized customer targeting strategies by building calibrated **predictive models** and providing curated call lists for sales teams. Led A/B testing to validate uplift, achieving a **30% increase in conversion rates**, higher annual contract values, and improved sales efficiency.
- Developed and deployed an **image-tagging system** that improved tag **F1-score** from 0.51 to 0.81 (+59%) across 200+ property-photo categories, enhancing listing accuracy and enriching AI-powered search and personalization.
- Designed and productionized **text embedding pipelines** for multimodal semantic search, enabling natural-language home discovery and informing the platform's transition toward AI-driven tagging and retrieval systems.
- Performed ad-hoc analytical deep-dives and communicated insights to executives and stakeholders to guide strategic decisions.
- Conducted **code and design reviews** to maintain a robust codebase, encourage best practices, and foster team-wide knowledge sharing.

Python SQL PyTorch Metaflow AWS Pinecone OpenAI Mistral Pydantic Gemini Hugging Face ONNX

Data Scientist II • [Intact Labs](#) • Toronto, Canada *Jun. 2020 – Jun. 2021*
Insurance leader building ML products for retention and growth.

- Built and productionized ML pipelines predicting business retention, optimizing profit, policy growth, and premium growth objectives.
- Monitored and diagnosed models in production to detect and address **ML drift**, ensuring stable prediction quality over time.
- Reviewed pull requests from teammates and provided actionable feedback to improve reliability, prevent bugs, and strengthen team knowledge sharing.

Python Scikit-learn Docker AWS GitHub

Data Scientist • Stradigi AI • Montreal, Canada

Nov. 2018 – May 2020

Enterprise AI platform enabling secure, production ML delivery.

- Designed and developed reusable **data preprocessing and transformation libraries** in Python for efficient data cleaning and manipulation within Stradigi's AI platform.
- Built end-to-end **ML pipelines** spanning sentiment analysis, document classification, time-series forecasting, and object detection across classical and deep learning approaches.
- Devised and developed a **benchmarking framework** to analyze and optimize ML pipelines against competitors.
- Worked closely with researchers and AI developers to ensure production-quality software and reliable ML delivery.

Python PyTorch TensorFlow Scikit-learn Docker AWS

Data Science Intern • Unbounce • Vancouver, Canada

Apr. 2018 – Jun. 2018

Conversion optimization product for marketing and growth teams.

- Developed and automated data pipelines using Python to extract, clean, and transform data into model-ready components.
- Applied survival analysis modeling techniques in R to identify behaviors influencing customer retention.
- Built clear visualizations and summaries to communicate findings to non-technical stakeholders.

Python Pandas R Survival Analysis GitHub

Software Developer • IMS Health (IQVIA) • Auckland, New Zealand

Sep. 2013 – Feb. 2016

Global healthcare analytics and technology company.

- Developed and updated functionalities for Nexxus Mobile Intelligence, a CRM solution supporting customer engagement in the life sciences industry.
- Performed maintenance and bug fixing on the existing codebase to improve reliability and ensure production-quality software.

C# JavaScript Angular.js Oracle

EDUCATION

University of British Columbia

Master of Data Science

Aug. 2017 – Jun. 2018

Vancouver, Canada

University of Waikato

Bachelor of Engineering (First Class Honors) – Software Engineering

Aug. 2009 – Aug. 2013

New Zealand

For more experience, projects and blog posts, please visit sarora.github.io.